MIL-D-18072D
19 August 1983
SUPERSEDING
MIL-D-18072C
26 FEBRUARY 1971

MILITARY SPECIFICATION

DRILLING MACHINES, UPRIGHT, SENSITIVE, BENCH TYPE, SINGLE SPINDLE, 5/16 INCH AND UNDER, HAND FEED

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric motor driven, bench mounted, highly sensitive, single spindle, hand feed, upright drilling machines with equipment as described herein.

1.2 Classification. The drilling machines covered by this specification shall be of the following types, classes, style and sizes, as specified (see 6.2.1).

Type I - Ultra Sensitive
Class 1 - Fixed Head

Size	Drill Capacity
1	0.001 to 0.0625 inch (.025mm to 1.5875mm)
2	0.002 to 0.125 inch (.051mm to 3.175mm)

Type II - Super Sensitive
Class 1 - Fixed Head

Size	Drill Capacity
1	0.0.35 to 0.250 inch (0.3429mm to 6.35mm)

Class 2 - Adjustable Head
Style A - 2 Speed Drive, Continuously Variable Speed Range

Size	Drill Capacity
1	0.004 to 0.3125 inch (0.1016mm to 7.9375mm)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, Memphis, TN 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Style B - Single Speed Drive, Four-increment Range

Size	Drill Capacity
1	0.025 to 0.3125 inch (0.635mm to 7.9375mm)

Type III - Precision

Class 1 - Fixed Head

Size	Drill Capacity
1	0.004 to 0.250 inch (.1016mm to 6.35mm) 0.0135 to 0.250 inch 0.3969 to 6.35mm)

Class 2 - Adjustable Head

Size	Drill Capacity
1	0.001 to 0.3125 inch (0.25mm to 7.9375 mm)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U.S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910)

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(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B5.10 - Machine Tapers, Self-holding and Steel Taper Series
B5.1 - T-Slots, Their Bolts, Nuts, Tongues, and Cutters
ANSI/IEEE 268 - Metric Practice
B11.8 - Construction, Care and Use of Drilling, Milling and Boring Machines, Safety Requirements for the

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - Commercial Packaging, Standard Practice for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG1 - Motors and Generators
ICS - Industrial Controls and Systems

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, NW, Washington, DC 20037.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish a sample unit for first article inspection and approval (see 6.3).

3.2 Design. The machine shall be new and one of the manufacturer's current models capable of drilling workpieces within its specified capacity in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

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3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in the U.S. Customary System of Units (US). These measurements may be converted to the International System of Units (SI) through the use of the conversion factors and methods specified in ANSI/IEEE268.

3.2.1.1 Measuring and indicating device calibrations. When specified (see 6.2.1), measuring and indicating devices such as feed screw dials, scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and all other similar devices shall be graduated in the specified system (US or SI) or measurements. Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.2 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI System of Measurements. When a dual US and SI system is furnished, both feed dials shall have independent zero adjustments and both dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Controls. All operating controls shall be located convenient to the operator at his normal work station and shall conform to NEMA Standard ICS.

3.2.4 Safety and health requirements. Covers, guards or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with operation of the machine. The safety devices shall prevent unintentional contact with the guarded part and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms and assemblies furnished on the machine, whether specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4). In addition, the machine shall comply with all requirements of ANSI B11.8 that are designated therein as the responsibility of the machine manufacturer. In the event of a conflict between the requirements of the OSHA and ANSI Standards, the requirements of OSHA shall apply.

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3.2.5 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Each recirculating system shall include a filter which is cleanable or replaceable. Each lubrication reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.3 Construction. The drilling machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No process such as peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts and other fasteners shall be installed in a manner that prevents change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the drilling machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All dials that indicate stock removal or tool movement shall be graduated in increments not larger than .001 inch or .025mm. Dial diameters shall be such that graduations may be read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a non-glare background.

3.3.8 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

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3.3.9 Gears. All gears shall be machined from a material suitable for the intended purpose. All gears shall be heat treated by a process that will impart the hardness and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature wear.

3.4 Components. Each machine shall consist essentially of a base, column, drill head and spindle assembly, worktable, driving mechanism and operating controls.

3.4.1 Base. The base shall be constructed of iron or iron alloy reinforced for rigidity and shall have a machined working surface which shall not deviate from a plane surface by more than 0.001 inch. Not less than two hold down lugs shall be provided for bench mounting. When T-slots are to be furnished, the number, direction, and location shall be as specified (see 6.2.1). Machined T-slots shall conform to ANSI B5.1. Types I and III machines shall be furnished with a trough type base. Trough type bases shall be provided with a coolant trough extending completely around the periphery of the work surface with provisions for a drain connection. Base dimensions shall be in accordance with table I. Bases for Type I, Type II Class 1, and Type III Class 1 machines shall be designed to accept the driving motor.

3.4.2 Column. The column shall be of cast iron or steel and shall be securely mounted to the base. The machine columns shall be of cylindrical cross section, ground to a finish not rougher than 16 microinches, and arranged to receive and support the drill head and spindle assembly, the table, and for Class 2 machines, the motor. Column deflection for all types shall not exceed 0.0005 inch in 6 inches of column length. Deflection shall be measured with the machine operating with its maximum rated drill capacity using recommended thrust.

3.4.3 Table. The table shall be of cast iron or iron alloy sufficiently reinforced to form a non-deflecting member. Tables shall have machined work surfaces which shall not deviate from a plane surface by more than 0.001 inch. The table shall be arranged for horizontal and vertical adjustment, shall be provided with a positive locking arrangement, and shall be readily removable. When T-slots are to be furnished, the number direction and location shall be as specified (see 6.2.1). Table shall be rectangular in shape except for Type I, Class 1, Size 1 and Type III, Class 2, which shall be round. The Type II, Class 2 and Type III, Class 1 machines shall be equipped with a combination round non-tilting table, designed to fit into the rectangular table.

3.4.4 Drill head. The drill head shall be accurately machined and finished and shall be rigidly supported by the column. When pulley drive is used, the drill head shall be arranged to contain and support the spindle pulley with quill and bearings, to drive and laterally support the spindle. Drill heads of the Type II, Class 2 and Type III, Class 2 machines shall swing radially on the column and Type I and Type III, Class 1 machines shall be fixed. Vertical adjustment of Type II, Class 2 and Type III, Class 2 machines shall be by means of handcrank, handwheel or by hand, with provisions for locking at any position without damage to the column.

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3.4.5 Spindle. The spindle shall be of tool steel, machined, hardened and finish ground to not rougher than 16 microinches. The spindle nose shall be hardened to at least Rockwell C55 and the keyway area to at least Rockwell C45. The spindle shall be mounted in precision type ball bearings arranged to take both radial and axial loads. Provisions shall be made for the spindle to return to its uppermost position through a universal ratchet adjustable spring, or counterweight. The spindle pully driving key shall be hardened to not less than Rockwell C35. When a drill chuck is not furnished as an integral part of the spindle or drive shaft, sockets of the spindle shall be tapered to conform to ANSI B5.10.

3.4.6 Depth stop. Each machine shall be provided with a depth stop equipped with a positive locking device to prevent change in adjustment during operation.

3.4.7 Chucks. Chucks shall be balanced, three jawed, and accurately fitted to the spindle, and shall be either key-operated or keyless as specified (see 6.2.1). Each key-operated chuck shall be provided with a key. Chucks shall be capable of holding straight shank drills covering the complete range for the applicable rated size machine.

3.4.8 Feed. Feed shall be accomplished by either a rack and pinion, a counter balance and lever system to the spindles or counter balance system to the table. In all cases, a sensitive feed shall always be obtained.

3.4.9 Drive. Drive shall be through belt and step pulleys or direct motor driven. The spindle shall be mounted in such a manner that driving torque only is transmitted to the spindle. All spindles and pulleys shall be dynamically balanced to avoid excessive vibration. When a belt and pulley is used as driving medium, provision shall be made for adjusting belt tension.

3.4.10 Illuminating lamp. An illuminating lamp shall be furnished for Type I, Type II and Type III, Class 2 machines. It shall have a protective shield, be adjustable, and shall be permanently attached to the machine in such a manner as to allow light to be swung over the workpiece.

3.4.11 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a strainer or baffle which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant.

3.4.12 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to all requirements in this paragraph. The system shall conform to the requirements of NEMA-ICS. Each machine shall draw all of its electrical power from a single 115/230 volt, 1 phase, 60/50 Hz circuit. The machine shall be initially wired for operation on 115 volts. An identified terminal for grounding the machine when installed, shall be mounted in or near the disconnect switch.

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3.4.12.1 Motors. Motors shall have ball or roller bearings. Continuous duty horsepower ratings shall conform to the requirements of table I. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall be energy efficient. Motor characteristics shall conform to NEMA Standard MG-1.

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TABLE I - Sizes and capacities.

CHARACTERISTIC	REQUIREMENT							
	TYPE I		TYPE II			TYPE III		
	Class I		Class I	Class II		Class I		Class II
	Size 1	Size 2	Size 1	Style A Size 1	Style B Size 1	Size 1	Size 2	Size 1
Capacity, inches	.001 to .0625	.002 to .0125	.0135 to 0.250	.004 to .3125	.025 to .3125	.004 to .250	.0135 to .250	.001 to .3125
Swing, inches	16	21 1/2	6	10	11-3/4	6-1/2	6	10
Vertical travel of head, inches	4-1/8	5-3/8	-	6	8	-	-	6
Spindle travel, inches	15/16	1-3/4	2-1/4	2-1/2	2-3/8	2-1/4	2-1/4	-
Distance chuck to base (Max), inches	7	8-3/4	8-1/2	14	10-7/8	13	14	-
Distance chuck to round table (Max), inches	2	-	-	10	8-3/4	8-1/2	9-1/2	4
Distance chuck to rectangular table, inches	-	4-1/4	4-3/4	9-1/2	8	8	9	-
Column diameter, inches	-	-	1-1/2	2	1-1/2	1-1/2	1-1/2	2
Spindle diameter, inches	1/4	3/8	7/16	9/16	9/16	9/16	9/16	9/16
Dimensions of base, width and length, inches	25 x 33	26 x 39	10 x 20	10 x 18	10 x 18	7 x 18	7 x 19	5 x 8
Table feed, inches vertical	-	2-1/2	-	-	-	-	-	1-5/8
2 directions horizontal	-	2	-	-	-	-	-	1
Spindle speeds, number	Infinitely variable	Infinitely variable	4	Infinitely variable	4	4	4	Infinitely variable
Speed range, revolutions per minute	275 to 8125	1300 to 3400	750 to 6000	840 to 2700	750 to 6000	800 to 7000	2450 to 6000	0 to 17,000
Motor, horsepower	1/16	1/4	1/4	1/3	1/4	1/4	1/6	1/16

Requirements are not less than shown. Where a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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3.5 Sizes and capacities. The sizes and capacities of the machine shall conform to the requirements of table I.

3.6 Performance. Each machine shall be capable of drilling five through holes, of its maximum rated drill size in accordance with 4.5.3.

3.7 Aligment tolerances. Each machine shall meet alignment accuracies of table II in accordance with paragraph 4.5.2.

TABLE II. Alignment accuracies.

CHARACTERISTIC	MAXIMUM ALLOWABLE DEVIATION
Spindle Taper Runout TIR at 1 inch from face	.001 inch
Spindle Square with worktable in 180° sweep at 6 inch radius	.002 inch
Quill travel square to table spindle incline outward from column in 12 inches	.0015

3.8 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. One set of hand tools normally furnished by the manufacturer.

b. One drill vise having steel jaws approximately 3 inches wide, 3 inch opening and 1-1/2 inches deep.

c. One extra drive belt, if applicable.

d. For type I machine, necessary collets or a chuck, to enable the machine to cover the specified drill range for the size machine required.

e. For type III, class 2 machines, either a magnetic chuck kit or a wire drill chuck to cover required specified drill range.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2.1).

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3.10 Additional requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified (see 6.2.1).

3.11 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall comply with the requirement of MIL-STD-461.

3.12 Fungus control. When required (see 6.2.1), fungus proofing shall be as specified.

3.13 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each drilling machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The chart or plate shall contain the following information:

Points of lubricant application
Service interval
Type of lubricant
Viscosity

3.14 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion-resistant metal nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.15 Technical data. Technical data shall be furnished as specified (see 6.2.1).

3.16 Workmanship. Workmanship of the drilling machine and accessories shall be commensurate with the requirements specified herein, and shall be of a quality prevailing among manufacturers normally producing equipment of the type specified herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. When first article approval is required under 3.1, first article inspection shall be performed. Unless otherwise specified (see 6.2.1), first article inspection shall comprise the examination in 4.4 and all tests in 4.5. Failure of the item to pass any examination or test shall be cause for disapproval of the first article.

4.3 Quality conformance inspection. Each item shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.4, the tests in 4.5 through 4.5.3 and the inspection in 4.6. Failure of the item to pass any examination, test, or inspection shall be cause for rejection.

4.4 Examination. The drilling machine and equipment shall be examined to determine compliance with the requirements in 3.2 through 3.5 and 3.8 through 3.16.

4.5 Tests.

4.5.1 Operational test. The drilling machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.5.2 Alignment accuracy tests. Each drilling machine shall be tested to determine that it complies with the alignment accuracy requirements of table II (see 3.7).

4.5.3 Performance tests. Each machine shall be tested by drilling five through holes, of the maximum drill size in the range covered by the machine required, in annealed chrome vanadium steel 0.025 inch thick, in not more than one minute.

4.5.5 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.11.

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4.6 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Packaging - packing and marking. Unless otherwise specified, packaging, packing and marking shall be in accordance with ASTM Standard D 3951. When required, level A packaging and level A or B packing and marking shall be in accordance with the applicable requirements of MIL-M-18058. The required levels and special marking shall be as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The drilling machines covered by this specification are intended for use in any toolroom, production shop or laboratory, where precise and accurate drilling of small holes is performed.

6.2 Ordering data.

6.2.1 Acquisition documents. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Type, class, size, and style required (see 1.2).
c. When a first article is required for inspection and approval (see 3.1, 4.2 and 6.3).
d. If machine is required to be configured in a specific measurement system (US or SI) state required system (see 3.2.1).
e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI) state required system (see 3.2.1.1).
f. Dual calibrations (US or SI); if required (see 3.2.1.2).
g. Specify additional safety and health requirements (see 3.2.4).
h. Painting, if different (see 3.3.5).
i. Number, direction and location of T-slots in base of table (see 3.4.1 and 3.4.3).
j. Specify if chuck is key or keyless operated (see 3.4.7).
k. Electrical system, if different (see 3.4.12).
l. Type motor enclosure, if different (see 3.4.12.1).
m. Standard equipment, if different (see 3.8).
n. Optional equipment, if required; fully describe required optional equipment (see 3.9).
o. Specify additional features required (see 3.10).
p. Electromagnetic interference control, if required (see 3.11).
q. Type fungus proofing, if required (see 3.12).
r. Lubrication chart or plate, if different (see 3.13).
s. Nameplate, if different (see 3.14).
t. First article inspection, if different (see 4.2).
u. Quality conformance inspection, if different (see 4.3).
v. Specific packaging, packing and special marking, if required (see 5.1). (Specify level A packaging, specify level A or b packing and specify special marking if required.)

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6.2.2 Contract data requirements. Required technical data, such as operators manuals, parts lists, wiring diagrams, and other instructions for operation and maintenance as identified on a numbered DD Form 1664 should be specified on a DD Form 1423 incorporated into the contract.

6.3 First article. When first article approval is required under 3.1, the solicitation should include complete information concerning examination, testing and approval.

6.4 Safety and health requirements. Paragraph 3.2.4 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, heat, etc. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment) the procuring agency is advised to analyze the existing hazards levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine to its future operating environment. If specific point-of-operation guarding is required, the procuring agency should specify the exact configuration of the guard required as in most cases, the guard configuration is dependent on the size and configuration of the workpieces that will be machined. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1(g).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review Activities:
Army - AL
Navy - SH
Air Force - 84
DLA - GS

User Activities
Army - SM
Navy - YD, MC

Preparing Activity:
DLA - IP

Project Number:
3413-0130

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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